



Three Routine Intraoperative Variables Predict Vasoplegia at Separation from Cardiopulmonary Bypass

No detectable loss of discrimination against a 38-parameter machine learning model in 704 adult cardiac surgery patients

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KEY FINDINGS

Three numbers already on every perfusion record predict vasoplegia at the moment of separation as well as a 38-parameter machine learning model.

0.858 vs 0.855

AUROC of 3 routine variables vs the 38-parameter elastic net (difference +0.003, 95% CI -0.020 to 0.027)

+0.08 to +0.09

AUROC added by the minute-level flowsheet to the same algorithms on static data (Holm-adjusted p = 0.010)

0.84 vs 0.57

Among 438 patients not yet hypotensive at separation: three-variable model vs terminal MAP alone

256 vs 3,209

Patients needed (Riley) for a pre-specified 3- vs a 38-parameter model; this cohort has 704

1 Background

Normal tone → **Vasoplegia**

Vasoplegia is distributive shock after cardiopulmonary bypass (CPB). Vascular tone collapses and arterial pressure falls while the heart itself pumps normally.

Treatment escalates through vasopressors to rescue agents such as methylene blue, hydroxocobalamin, and angiotensin II.

pressure holds → pressure falls

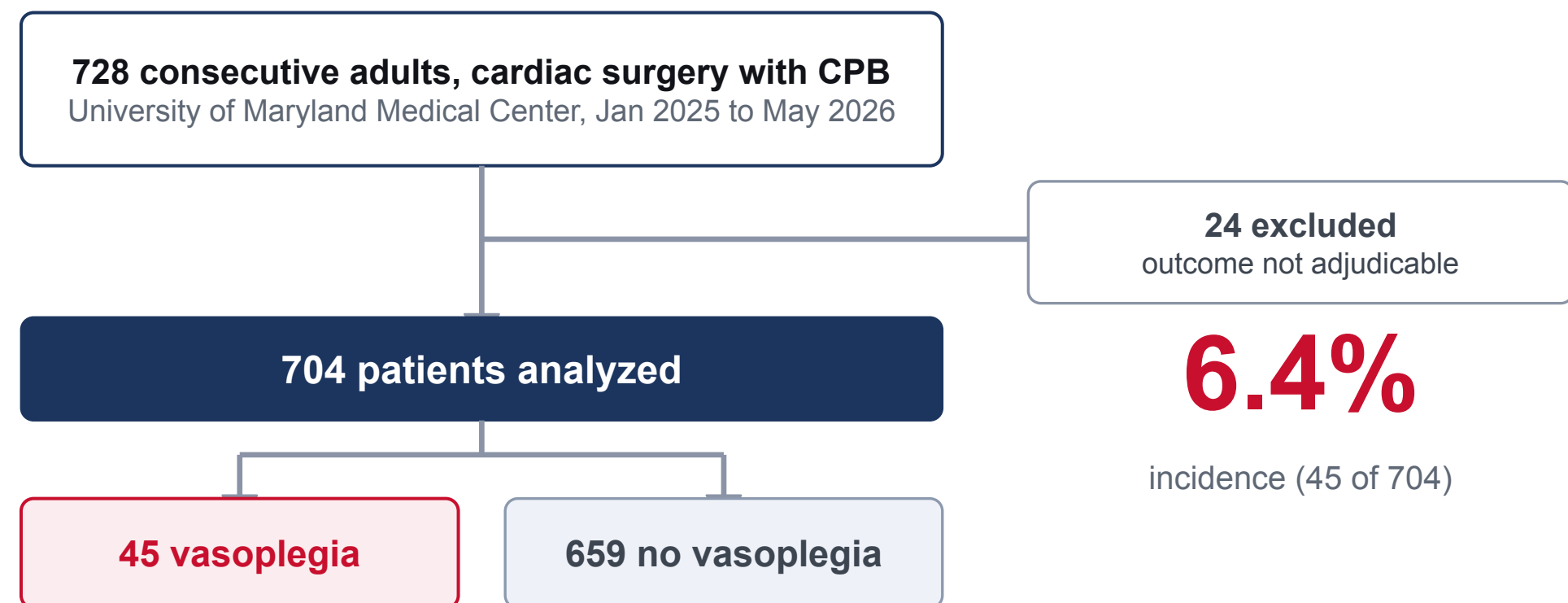
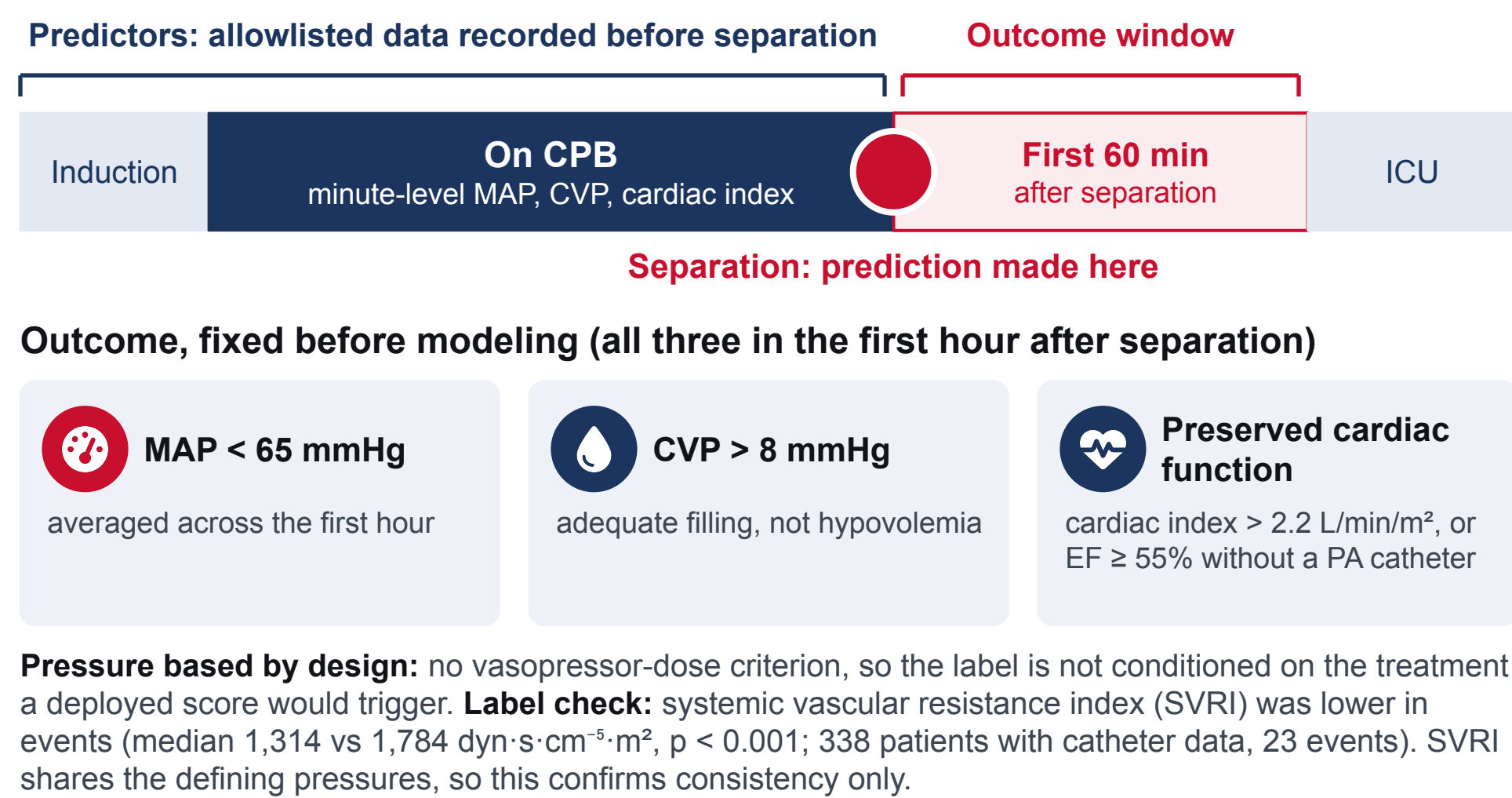
- Reported incidence spans 2.5% to 66%, largely because definitions differ [1, 2].
- Associated with longer intensive care stays and higher mortality [3].
- Usually recognized only after vasopressor needs escalate, although earlier rescue therapy is associated with better survival [4].

Where vasoplegia risk tools operate



Aim Early on-bypass hypotension was linked to vasoplegia with a single variable in 2009 [7], but no model has measured what the **minute-level intraoperative record** adds at the moment of separation, or how much model complexity a cohort with 45 events can support.

2 Study Population

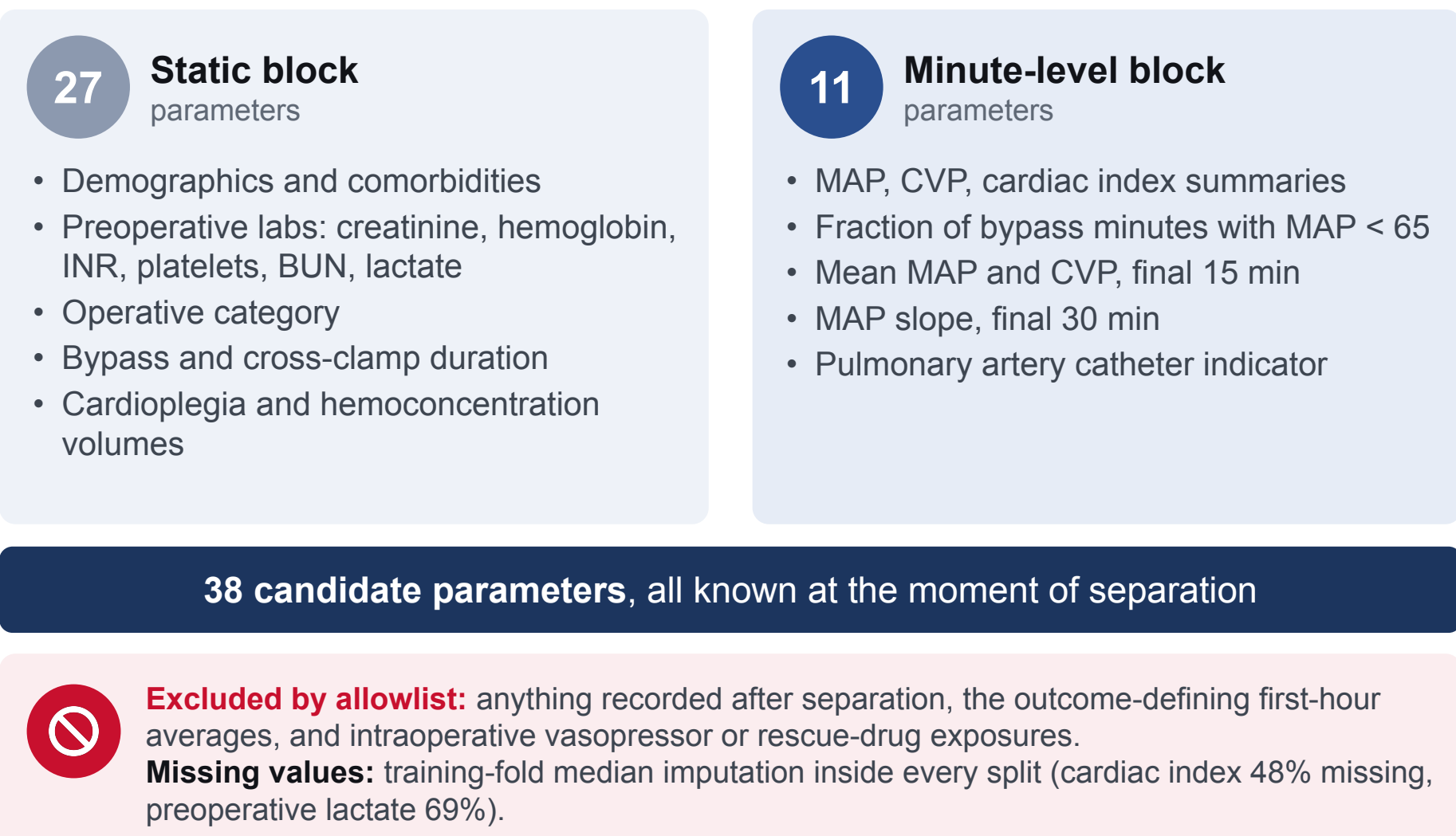


Cohort characteristics by outcome

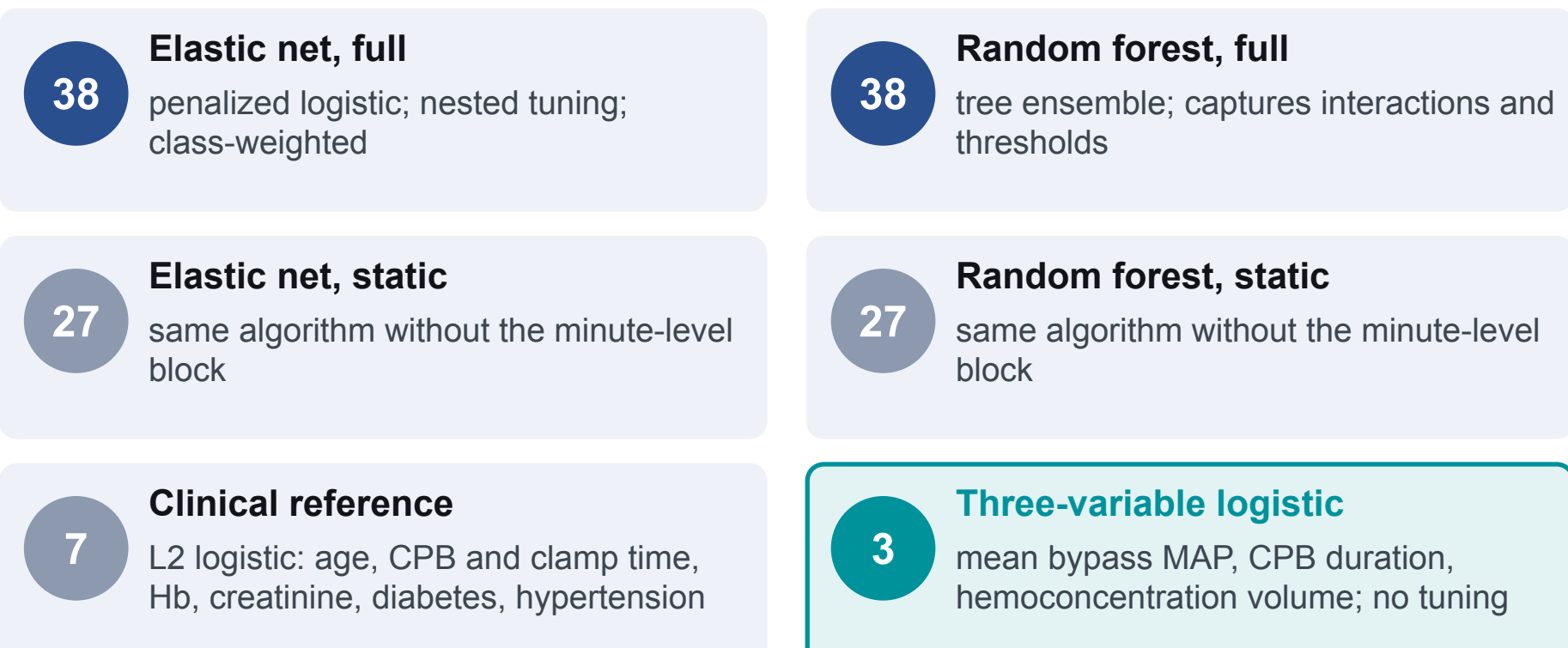
Characteristic	Vasoplegia (45)	None (659)	p
Age, years	64 [55, 70]	63 [55, 71]	0.75
Male sex	32 (71%)	451 (68%)	0.87
Diabetes	12 (27%)	175 (27%)	>0.99
CABG	16 (36%)	356 (54%)	0.020
Preop hemoglobin, g/dL	10.2 [8.9, 14.0]	13.1 [11.1, 14.4]	0.004
Preop creatinine, mg/dL	1.15 [0.91, 1.43]	0.99 [0.80, 1.20]	0.006
Preop INR	1.10 [1.00, 1.20]	1.00 [1.00, 1.10]	0.007
Aortic surgery	14 (31%)	111 (17%)	0.025
CPB duration, min	187 [146, 222]	116 [90, 156]	<0.001
Cross-clamp duration, min	122 [91, 159]	82 [59, 111]	<0.001
Mean MAP on bypass, mmHg	68.8 [65.2, 72.1]	74.8 [71.2, 78.3]	<0.001
Bypass minutes with MAP < 65	35% [24, 51]	22% [14, 31]	<0.001

Median [IQR] with Mann-Whitney p, or n (%) with Fisher exact p.

3 Data and Features

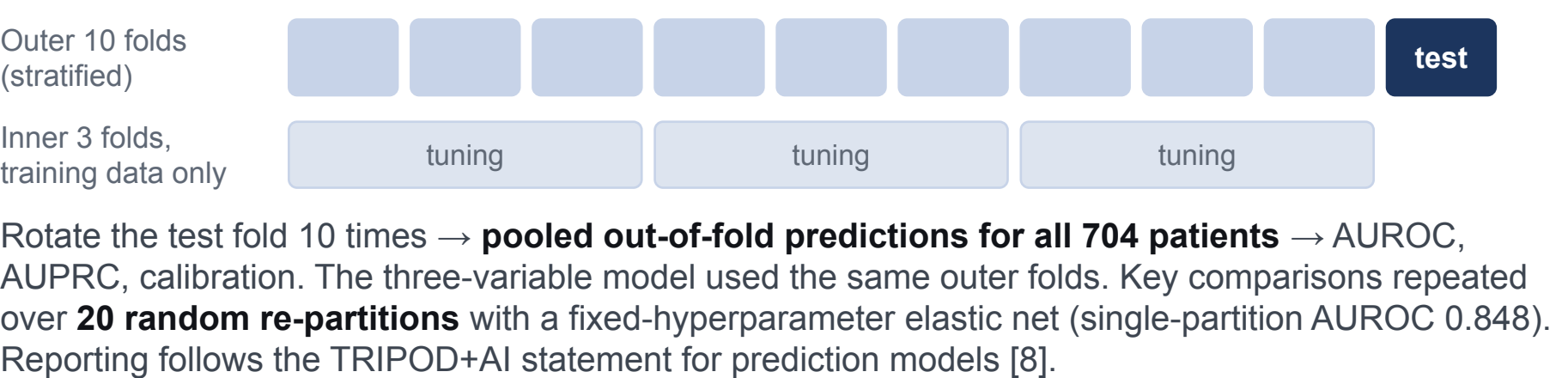


4 Model Development

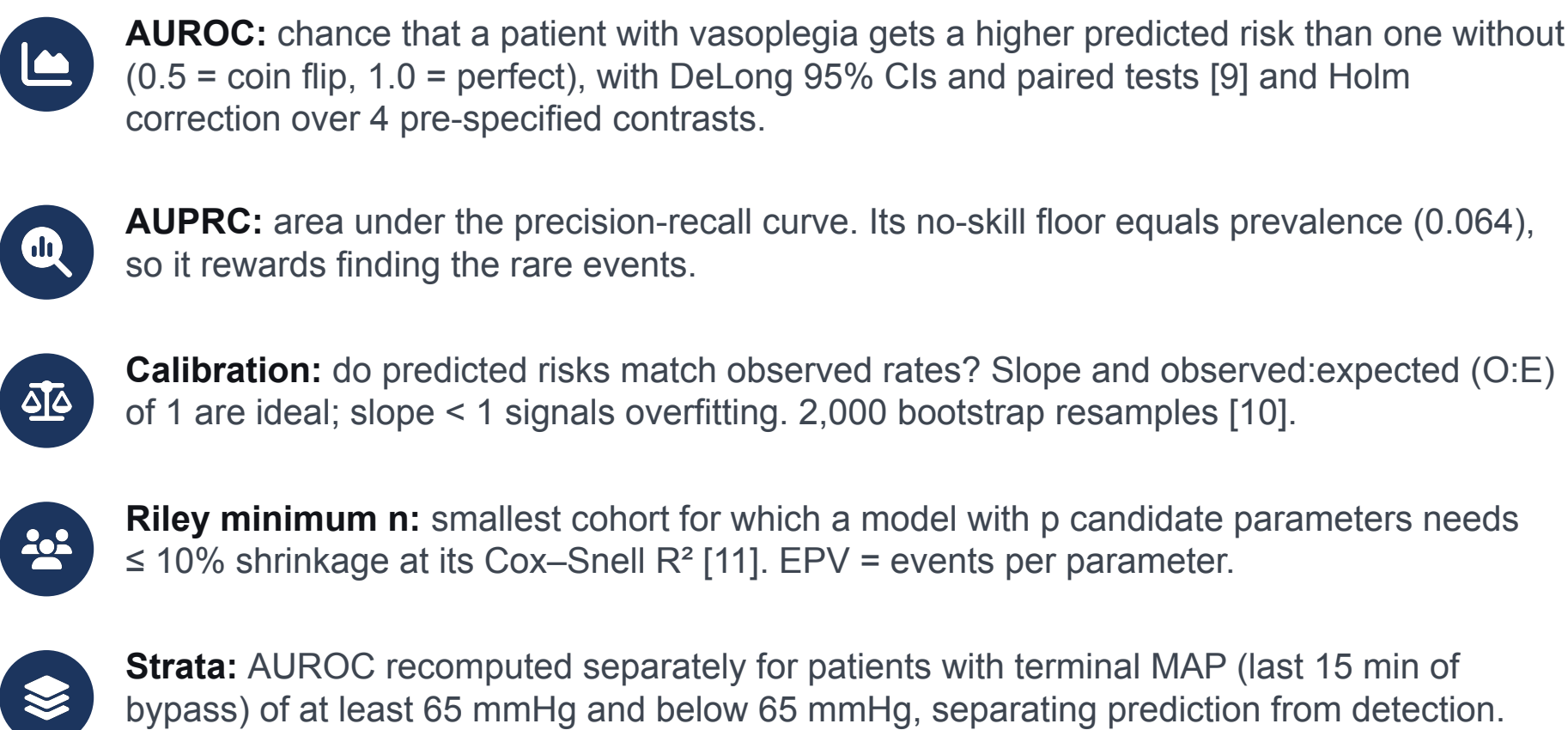


Three-variable model: the predictors the elastic net kept in all 10 folds with the largest standardized coefficients (one from the correlated pressure block), refit by unweighted maximum likelihood.

Nested cross-validation, scored strictly out of fold



5 Statistical Analysis

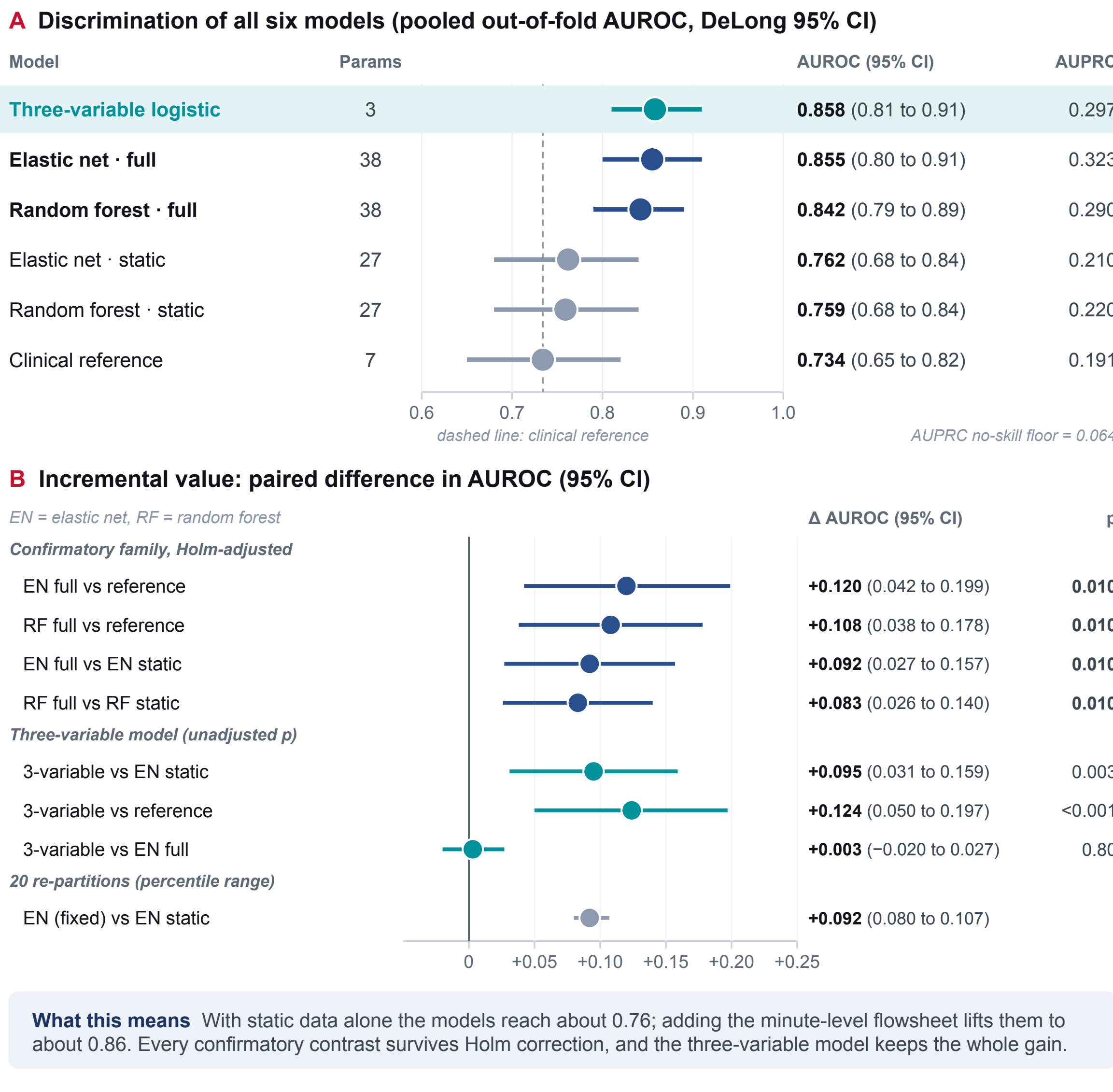


Pre-specified robustness checks (pooled out-of-fold AUROC, DeLong 95% CI)

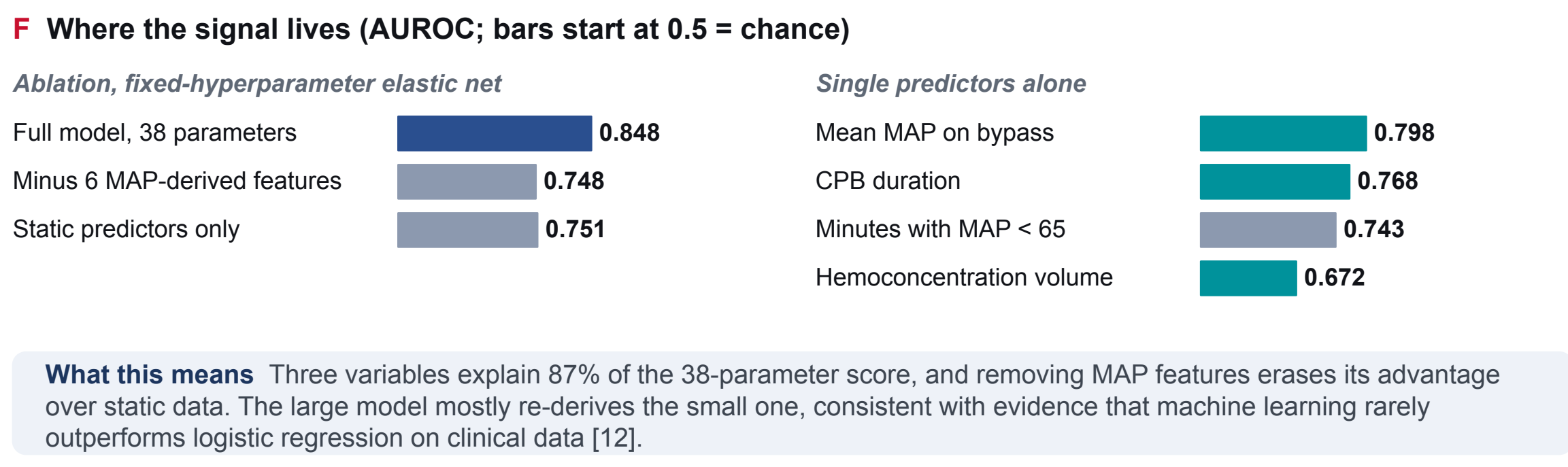
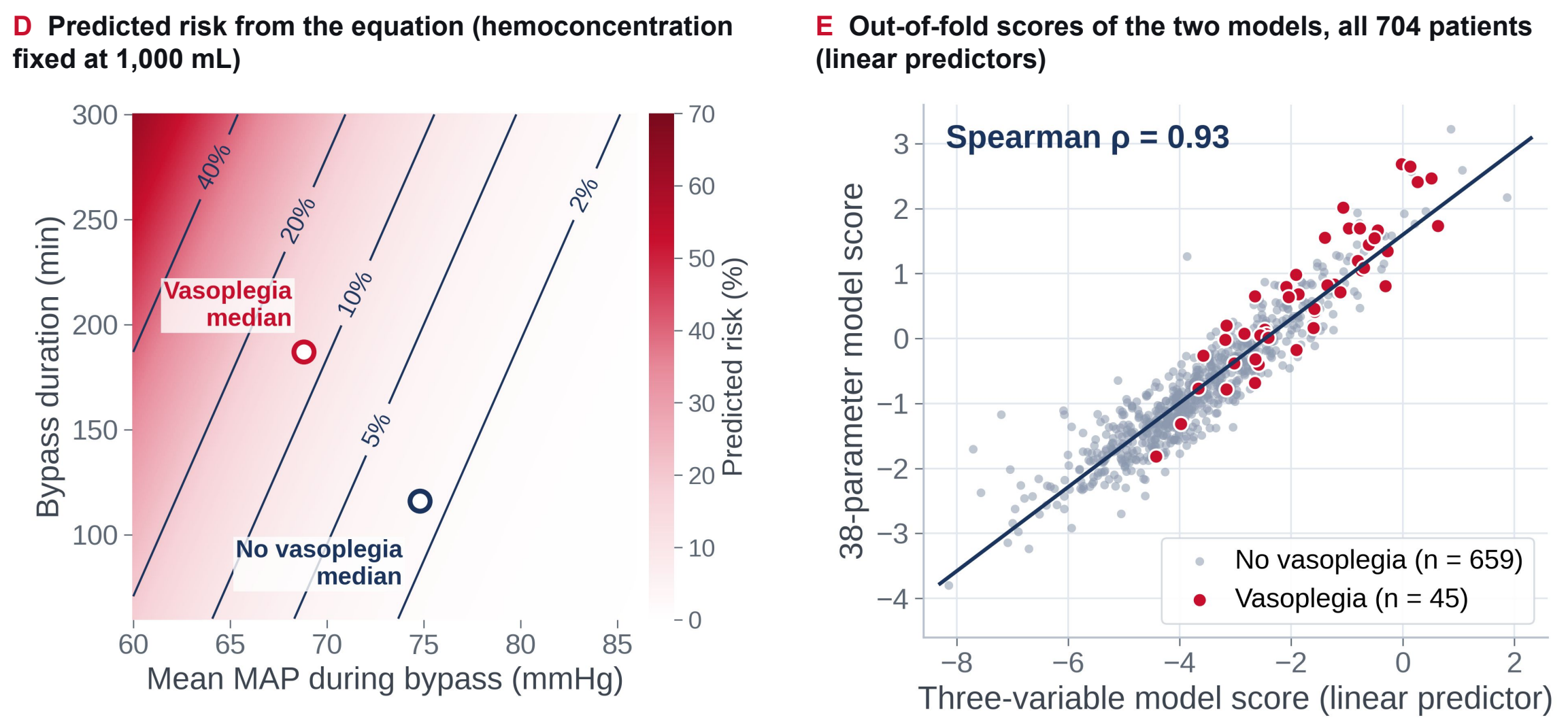
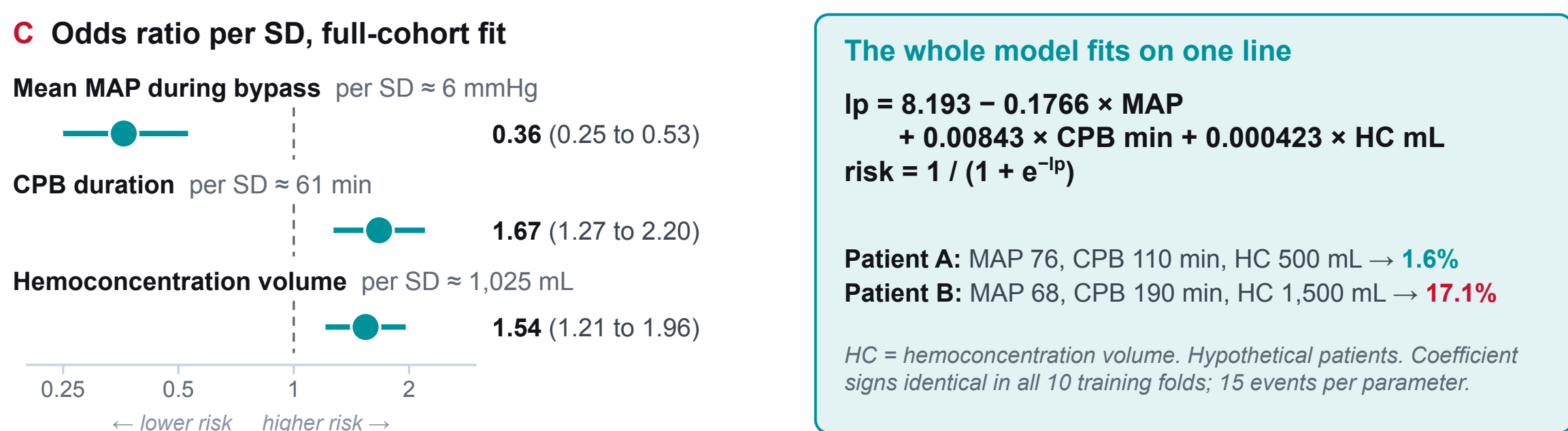
Terminal MAP in place of mean bypass MAP	0.849 (0.79 to 0.91)
Both MAP summaries (four variables)	0.874 (0.82 to 0.92)
Six stable predictors	0.869 (0.82 to 0.92)
Three-variable model with class weighting	0.855
Outcome at MAP < 60 mmHg (16 events), three-variable	0.876 (0.79 to 0.96); fold mean 0.89
Outcome at MAP < 60 mmHg, full elastic net	pooled 0.811; fold mean 0.88
Full model without catheter indicator	0.848
Full model with ambiguous-timing totals	0.864

Full-model rows use the fixed-hyperparameter elastic net, except the MAP < 60 row (nested-tuned), whose lower pooled value reflects probability-scale shifts across folds at 16 events rather than worse ranking.

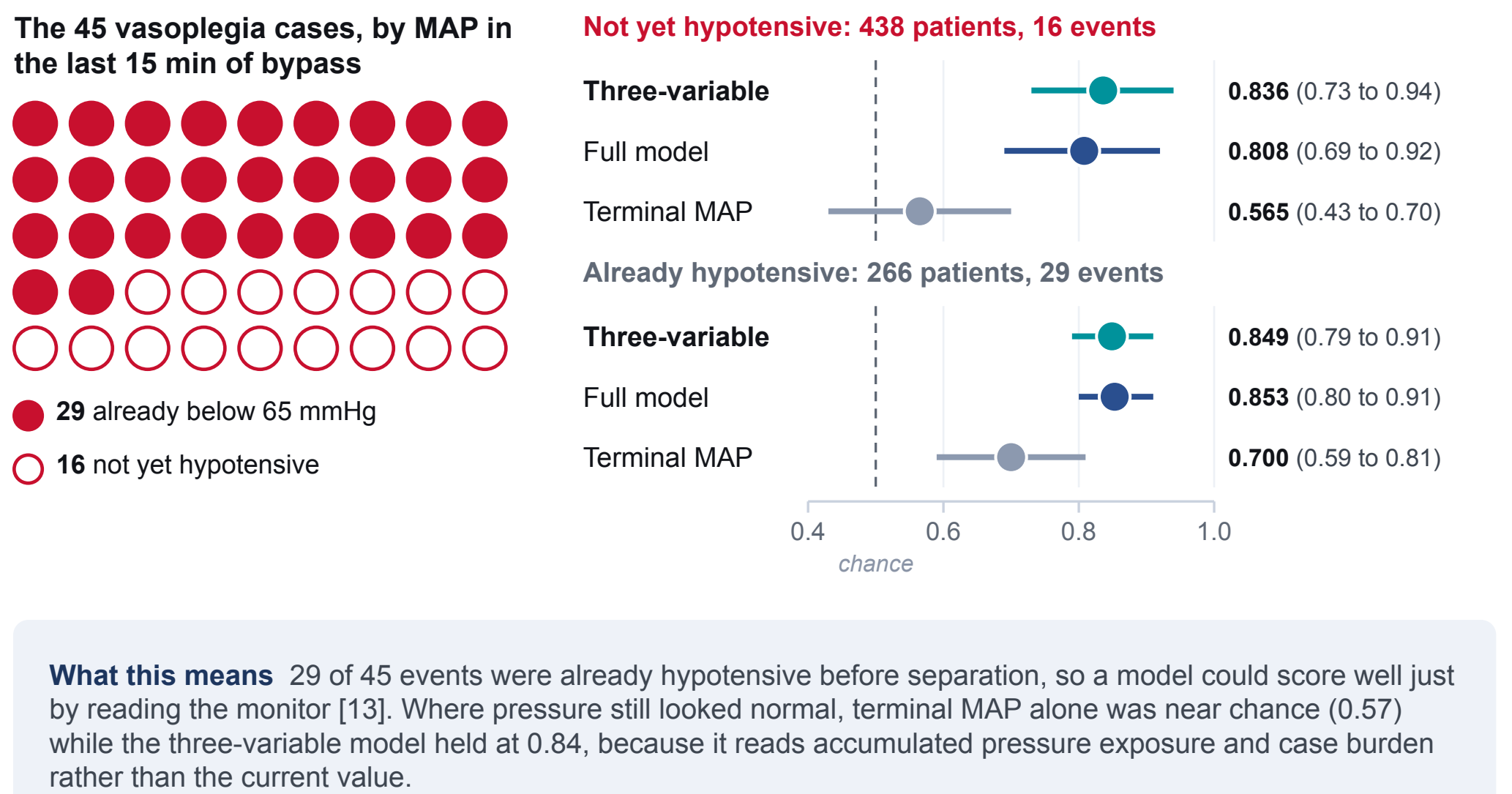
6 Model Performance



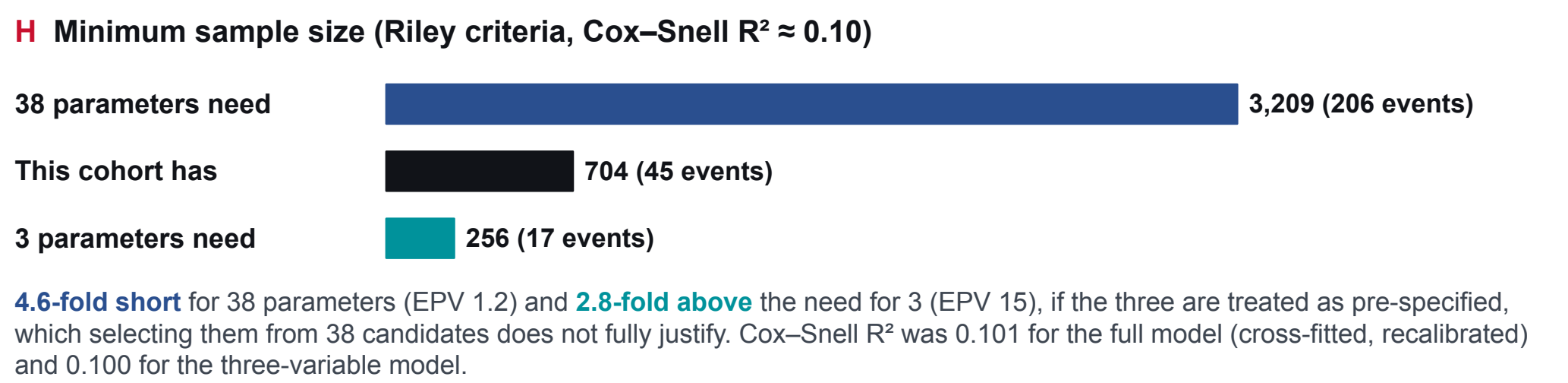
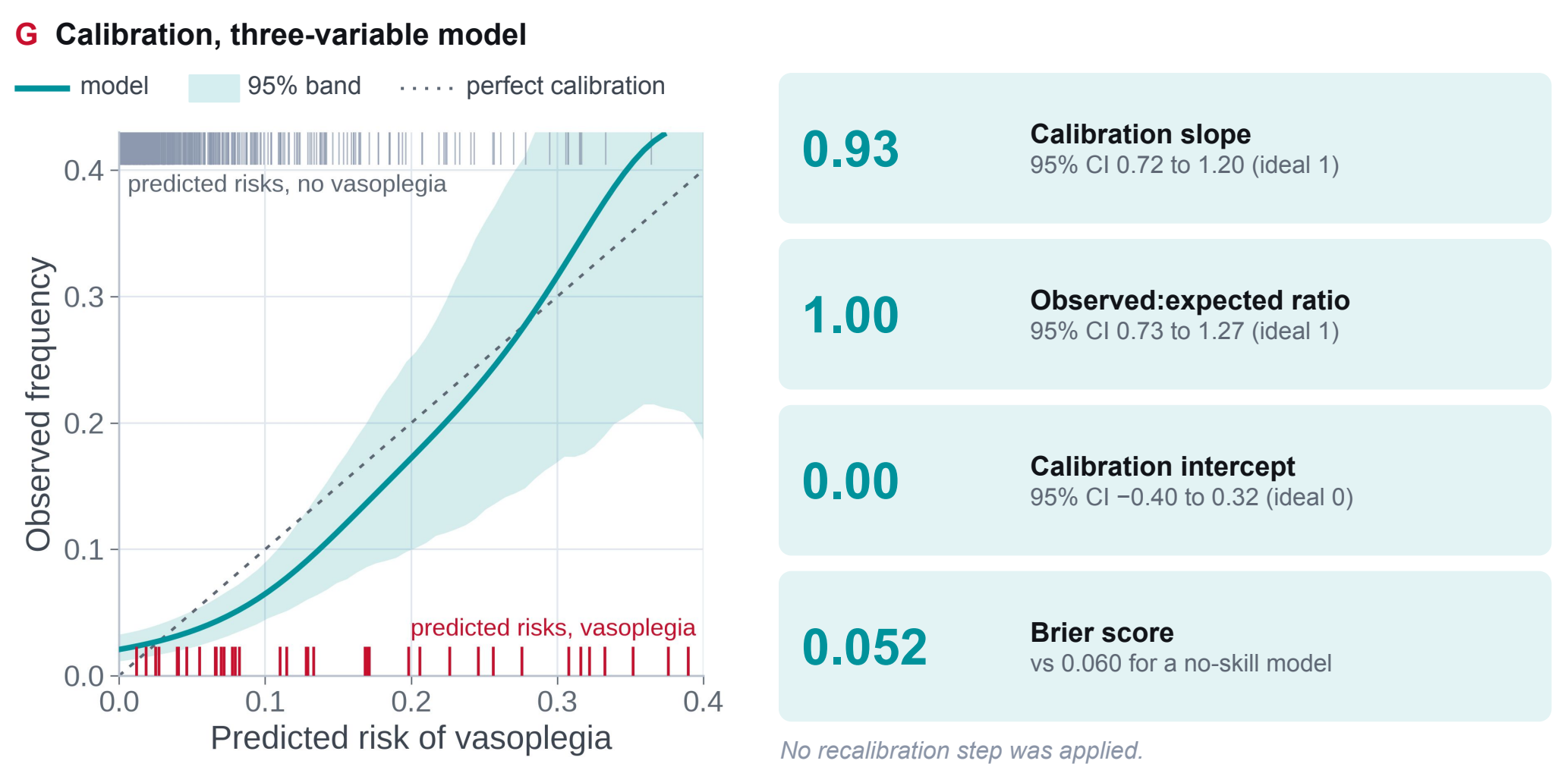
7 Three-Variable Model



8 Subgroup Analysis



9 Calibration and Sample Size



10 Limitations and Future Work

Selection: the three predictors were chosen once using all 704 patients before out-of-fold scoring, so 0.858 may be optimistic and the 3-parameter Riley figure assumes pre-specification; nested selection is planned.

Inference: p = 0.80 means no detectable difference, not formal equivalence; DeLong intervals on pooled predictions omit refitting and partition variance.

Scope: single-center retrospective study without external validation; 45 events overall, and only 16 among patients not yet hypotensive at separation.

Data: counts are provisional until a few screened records still under curation are resolved; cardiac index is missing in 48% and preoperative lactate in 69%.

Label: pressure based without a vasopressor criterion; at a 60 mmHg threshold only 16 events remain.



Expanded cohort: a 3- or 5-year total window would add about 787 (= 50 events) or 1,781 (= 114 events) new patients; the final size will follow a validation-specific calculation [15].

Conclusion Three numbers on every perfusion record, mean bypass MAP, bypass duration, and hemoconcentration volume, predicted vasoplegia at separation from CPB as well as a 38-parameter machine learning model, with calibrated risks and within Riley sample-size criteria for a three-parameter model. The next step is validating the frozen model in new patients.

References Acknowledgments: The Tensor Lab for Computational Medicine; UMSOM Office of Student Research.

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